

Applying Federated Transfer Learning to Small Molecule Datasets

Distributed Optimization for Machine Learning Final Presentation

ECE 5290

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The Pharma Data Silo Dilemma

Lead discovery: first stage of drug development. Involves searching through many compounds for one(s) with desired properties.

ML can speed this up! But currently has problems...

- Requires data on millions of compounds
- Experimental data is very expensive, thus scarce
- ...so good, highly labeled samples especially rare
- ...and pharma companies don't share the data they have

Could distributed learning be a solution?



Hurdles to Distributed Learning

So, can companies build better models together while still maintaining data privacy? *Note: lead generation is least expensive part of process, so makes some sense to collaborate on models.*

Problem: each institution will have data on different compounds, and with different experiments! How to learn one model from data with small overlap?

News | Published: 06 November 2025

Drugmakers share data to feed voracious foundation models

[Andrew Marshall](#)



Types of Distributed Learning



(a) Horizontal Federated Learning (b) Vertical Federated Learning (c) Federated Transfer Learning

From "Vertical Federated Learning: Concepts, Advances and Challenges" by Liu et al.

MELLODDY EU-funded project used FTL to build database from 2.6bn confidential data points from 10 of the world's largest pharma companies: Amgen, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, GSK, Janssen, Merck KGaA, Novartis, Servier.

Considering they sold off their website domain to an Italian casino while I was working on this project...I'm skeptical about their success.



Project Goals

The goals of this project:

- Identify a good biochemical data task for FTL testing
- Baseline with single-device and horizontal learning
- Preserve data silos with vertical learning on shared samples
- Preserve data silos while using all data with FTL



The Data: Moleculenet

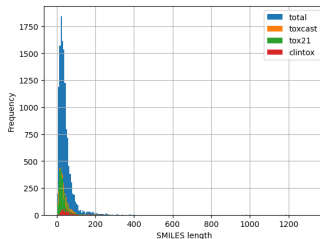
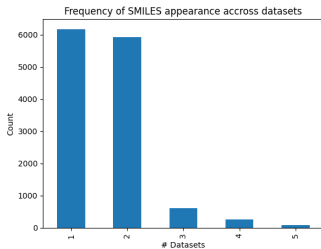
Moleculenet has small (1-8k samples) datasets for benchmarking scientific ML models in quantum mechanics, physics, biophysics, physiology...

SMILES to physiological classification task datasets:

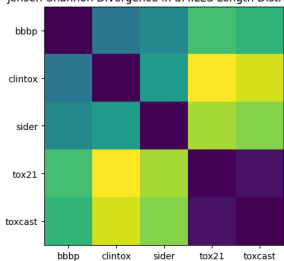
- **bbbp**: whether compound can be delivered to brain
- **clintox**: whether compound was toxic in clinical trial, and whether is FDA-approved
- **sider**: whether compound triggers side effects in various organ systems
- **tox21**: whether compound was positive for 12 preclinical toxicity screenings
- **toxcast**: whether compound was tested/positive for 600+ preclinical toxicity screenings



Data Exploration



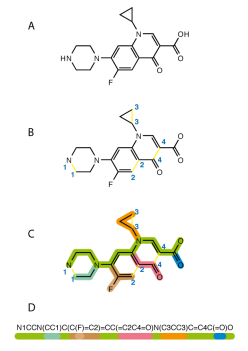
Jensen-Shannon Divergence in SMILES Length Distribution



SMILES

How do you turn a molecule into a sequence of LLM tokens?

- DNA, proteins: easy, already a sequence
- Small molecules: **SMILES** (Simplified Molecular Input Line Entry System)



People originally used GNNs for small molecules and preserved the structural aspect, but LLMs on SMILES sequences are more popular recently.

Sequence to Embedding to Prediction

SMILES

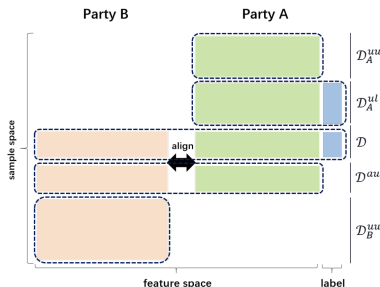
- Canonicalize and align
- **UME** (Universal Molecular Encoder, from Prescient Design)
- molecular embedding

Embeddings + other features from tox21, clintox

- Simple MLP (distributed training)
- FDA approval prediction



Vertical Federated Learning: FedBCD



From "Vertical Federated Learning: Concepts, Advances and Challenges" by Liu et al.

The data contains $\mathcal{D}$, $\mathcal{D}_B^{uu}$, $\mathcal{D}_A^{ul}$. For vertical training, we just want to use $\mathcal{D}$ and train with datasets A and B on different nodes. For FTL, we want the two nodes to also use $\mathcal{D}_B^{uu}$ and $\mathcal{D}_A^{ul}$.

But how do you do multi-node learning when only one node has the labels?



Vertical Federated Learning: FedBCD

Vertical federated learning objective:

$$\min_{\Theta} \mathcal{L}(\Theta; \mathcal{D}) \triangleq \frac{1}{N} \sum_{i=1}^N f(\theta_1, \dots, \theta_K; \mathcal{D}_i) + \lambda \sum_{k=1}^K \gamma(\theta_k)$$

Where $\mathcal{D}_i^k$ is the parts of sample i held by node k . If k is the labeled node, $\mathcal{D}_i^k \triangleq \{x_i^k, y_i^k\}$, else $\mathcal{D}_i^k \triangleq \{x_i^k\}$. f is loss function and γ is regularizer.

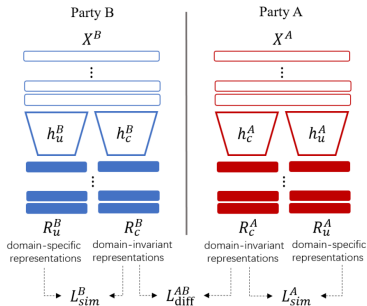
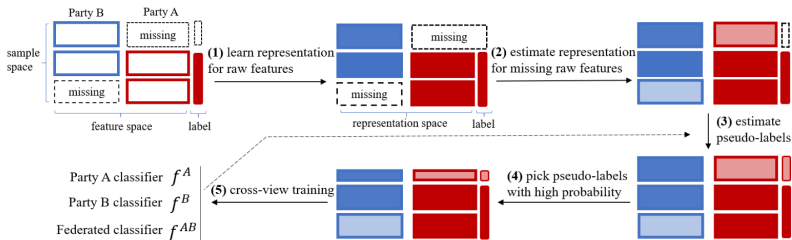
The parallel **FedBCD** method works by nodes exchanging local weights multiplied by the data points, $H_i^k = x_i^k \theta_k$, between rounds of multiple local gradient updates on batches $\mathcal{S}$. Each node uses its own weights and batch alongside the H of others to compute its gradient (can use stale $H_{-k}^{\mathcal{S}}$ for multiple local updates):

$$H_{-k}^{\mathcal{S}} := \{H_q^k(\theta_q, \mathcal{S}^q)\}_{q \neq k}.$$

$$g_k(\Theta; \mathcal{S}) = \nabla_k f(H_{-k}^{\mathcal{S}}, \theta_k; \mathcal{S}) + \lambda \nabla \gamma(\theta_k)$$



Federated Transfer Learning: FedCVT



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